Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18MT51]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

NON METALLIC MATERIALS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	a.	Distinguish between the metals and non-metals.	07
	b.	Illustrate the various flat glass manufacturing process with neat sketch.	07
2.	a.	Discuss the steps involved in slip casting process using ceramic powder with neat sketch.	07
	b.	Explain the following mechanical preparation methods of producing ceramic powder with a neat sketch: i. Ball milling, ii. Roller Crushing process.	07
3.	a.	How vacuum forming and match die process were utilized for manufacturing of polymer parts.	08
	b.	Explain the working principle of blow molding process with neat sketch.	06
		OR	
4.	a.	What is injection molding process and explain its working principle with neat a sketch.	08
	b.	Elucidate the mechanical properties of plastics with a neat sketch.	06
5.	a.	How CMCs parts are produced using directed oxidation process. Explain with neat sketch.	09
	b.	Explain the classification of composite based on geometry of reinforcement.	05
		OR	
6.	a.	How toughening takes place and fracture toughness improves in case of reinforced ceramics. Explain with neat sketch.	07
	b.	Explain the spray lay-up process with neat sketch for manufacturing of composite components.	07
7.	a.	Elucidate the hole making techniques; i. Wood pecker cycle and ii. Back-up plate technique for polymer matrix composites.	10
	b.	Illustrate the need for secondary processing of composites.	04
		OR	
8.	a.	Write down the difference between adhesive and mechanical Joints.	09
	b.	Explain the issues and challenges in adhesive bonding and bolting/fastening.	05